

Byeongjee Kang

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RESEARCH INTERESTS

Programming Languages Design and Implementation, Formal Methods, Distributed Systems, and Parallel Programming

EDUCATION

Carnegie Mellon University

Pittsburgh, USA

Ph.D. in Software Engineering (Advisor: Prof. Umut Acar)

Aug 2023 – Present

Pohang University of Science and Technology (POSTECH)

Pohang, Korea

Bachelor of Computer Science and Engineering/Mathematics (double major)

Feb 2018 – Feb 2023

- GPA : 4.26/4.3 (ranked 1st in class)
- Entered with *Program for highly Dedicated Students (PHDs)*, presidential scholarship and supporting program offered to one outstanding high school student in the year
- Graduated within 6 semesters (leave of absence for military duty during Jan 2020 – Dec 2021)

Korea Science Academy of KAIST (*High School for Science-Gifted Students*)

Busan, Korea

Graduated

Feb 2015 – Feb 2018

- GPA : 4.23/4.3 (ranked 1st in class)
- Funded by Korea Science Academy Fund Scholarship

PUBLICATION

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems, [Byeongjee Kang](#) and Kyungmin Bae, *Science of Computer Programming* 2024 (doi.org/10.1016/j.scico.2024.103097)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search, [Byeongjee Kang](#) and Kyungmin Bae, *International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS)* 2022 (doi.acm.org?doi=3563822.3568017)

PUBLICATION (UNDER REVIEW)

Mr3: An Execution Framework for Hive and Spark Sungwoo Park, Seunggon Namgung, [Byeongjee Kang](#), *Journal of Distributed and Parallel Computing* (dx.doi.org/10.2139/ssrn.4587845)

HONORS AND SCHOLARSHIPS

ILJU Fellowship (ILJU Academy and Culture Foundation)

Aug 2023 – July 2028

- Received total of USD 12,000 over eight semesters during graduate studies

Program for Highly Dedicated Students (POSTECH)

Feb 2018 – Feb 2023

- Presidential scholarship and supporting program in POSTECH offered to **one outstanding freshman** each year

Presidential Science Scholarship (Ministry of Education of Korea)

Feb 2018 – Feb 2023

- National science scholarship offered to top freshmen students in the field of science and technology

Global Leadership Program (CSE Department, POSTECH)

Sep 2019 – Present

- Scholarship and research support program for selected students in computer science department

Domun Scholarship (POSTECH)

Feb 2019

- Scholarship for freshmen with highest GPA

RESEARCH EXPERIENCES

POSTECH Software Verification Laboratory

Pohang, Korea

Research Intern - Advised by Prof. Kyungmin Bae

Feb 2022 – May 2023

- **Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search**
 - Designed and implemented search heuristic for narrowing-based symbolic reachability analysis of distributed systems
 - Paper accepted to *International Workshop on Formal Techniques for Safety-Critical Systems 2022*

POSTECH Programming Languages Laboratory

Pohang, Korea

Research Intern - Advised by Prof. Sungwoo Park

June 2018 – June 2019

- **MR3**: General purpose distributed execution framework written in Scala
 - Designed and implemented locality-aware task scheduler achieving higher throughput

WORK EXPERIENCES

Kodebox

Seoul, Korea

Software engineer as military service alternative

Jan 2020 – Dec 2021

- **CodeChain Foundry**: Open-source blockchain engine based on composable module system
 - Worked as software engineer
 - Implemented BLS signature aggregation scheme to optimize block seal size
 - Designed interface for modules to communicate with consensus engine
 - CodeChain Foundry is written in Rust
- **Zuzu**: Web-based captable management software
 - Worked as full-stack software engineer
 - Main features implemented include regression testing framework for document generation service, payment, and more
 - Tech Stacks: Django (Python), React (TypeScript), GraphQL, Kubernetes, Google Cloud Platform

TEACHING EXPERIENCES

Student Mentoring Program (CSE Department, POSTECH)

Feb 2022 – June 2022

- Served as tutor to teach Algorithm course

Intensive Research Program (POSTECH)

Jan 2019

- Served as mentor in research experience program for science high school students

Voluntary Teaching (POSTECH Volunteering Club)

Jan 2019 – Dec 2019, Sep 2022 – Dec 2022

- Served as voluntary teacher of math for unprivileged middle school students

EXCHANGE STUDENT & LANGUAGE PROGRAM

Summer Session Program

TU Berlin

Data science course

Aug 2019 — Sep 2019

English Program

University of New South Wales

English course

Dec 2017 — Jan 2018

AwesomeMath Summer Program

Cornell University

Mathematics problem solving program for gifted middle and high school students

June 2016 — July 2016

POSTECH Decentralized Autonomous Organization (PDAO) <i>Student association for blockchain development and research</i> • Served as mentor in blockchain core development team • Designed consensus algorithm to be used in PDAO	POSTECH <i>Feb 2022 – Present</i>
Intensive Research Program <i>Research experience program for science high school students - advised by Prof. Jae-Joon Kim</i> • Learned basic deep learning concepts and researched on techniques to reduce size of DNN	POSTECH <i>July 2017</i>
Research and Education (R&E) Program <i>Group research program for juniors - Advised by Dr. Youngchul Jeon</i> • Researched properties of <i>Abelian Ring</i>, a field of noncommutative ring theory • Presented result as school representative in Japan Super Science Fair (JSSF) 2016, at Ritsumeikan High School	Korea Science Academy of KAIST <i>Mar 2016 – Dec 2016</i>
Korea Young Physicists' Tournament (KYPT) <i>Team-based scientific competition for high school students</i> • Designed and performed experiments, and debated with other teams on the outcome • Won gold medal	The Korean Physical Society <i>June 2017 – Jan 2018</i>
POSCAT <i>Students club for algorithm study and competitive programming</i> • Committee member of Postech Programming Contest 2019	POSTECH <i>Feb 2018 – Dec 2019</i>
POSTECH Student Ambassador <i>Student-run POSTECH International PR group</i> • Voluntary service to support exchange students visiting POSTECH	International Relations, POSTECH <i>Feb 2018 – Dec 2018</i>
POBBA <i>Basketball club</i> • Served as club secretary during the year 2019	POSTECH <i>Feb 2018 – Present</i>

LANGUAGES

Korean (native), English (fluent)